SSC MTS 2024 question paper with answers

28 October 2024 · Shift 2

77 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 13 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 14 questions

Q.1

A batsman scores 152 runs in his 15th inning, which increases his average score by 5 runs. Find the average score per inning of the batsman after the 15th innings.

(A) 85
(B) 77
(C) 41
(D) 82

Q.3

The ratio of the I class and II class AC train coach fares between two stations is 5 : 3, and that of the number of passengers travelling between the two stations by I class and II classes AC coach is 3 : 10. If on a particular day ₹37,080 is collected from passengers travelling between the two stations, then find the amount collected from second class AC coach passengers.

(A) ₹22,047
(B) ₹24,720
(C) ₹20,247
(D) ₹27,420

Q.4

A shopkeeper offers 38% discount on a certain item. He announces an additional 15% discount to a customer for paying cash. A customer paid a cash amount of ₹2,108 for that item. What is the marked price (in ₹) of the item?

(A) 3,800
(B) 3,560
(C) 4,200
(D) 4,000

Q.7

Simplify the following expression. $(5 \times 5 + 5 - 5 \times 5 + 5 - 5 \times 5) + (5 \times 5) \div 5 (2 + 3)$

(A) 25
(B) 10
(C) 20
(D) 5

Q.8

Arun has taken some money as loan on simple interest for 6 years. The money lender levied interest at 4% per annum for the first 2 years, at 8% per annum for the next 3 years and at 12% per annum for the period beyond 5 years. If Arun clears his loan by paying ₹5,472, then what was the sum (in ₹) he had taken as loan?

(A) 3,800
(B) 3,040
(C) 12,436
(D) 6,840

Q.9

A shopkeeper sells an item at a profit of 25% and dishonestly uses a weight that is 30% less than the actual weight. Find his total profit percentage.

(A) 42%
(B) 78.57%
(C) 65.75%
(D) 23%

Q.10

A 735 m long train passes through a 225 m long tunnel in 48 sec. Find the speed of the train.

(A) 72 km/h
(B) 82 km/h
(C) 54 km/h
(D) 66 km/h

Q.12

In what ratio must a shopkeeper mix sugar at ₹30/kg and ₹32.5/kg so that by selling the mixture at ₹34.1/kg, he may get 10% profit?

(A) 2 : 1
(B) 3 : 2
(C) 5 : 4
(D) 4 : 3

Q.13

A shopkeeper sold the Refrigerator and the Geyser set together for ₹30,400, thereby making a profit of 25% on the Refrigerator and 10% on the Geyser. By selling them together for ₹30,700, he would have made a 10% profit on the Refrigerator and 25% on the Geyser. Now, if the Geyser is sold at 17% profit, then the selling price (in ₹) of the Geyser is:

(A) 16,380
(B) 12,130
(C) 17,510
(D) 14,380

Q.14

Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 8% per year rate of interest for 4 years.

(A) 960
(B) 900
(C) 1000
(D) 920

Q.15

The average weight of a family of five members (in kg) whose weights are 40 kg, 49 kg, 56 kg, 71 kg and 37 kg is:

(A) 51.6
(B) 52.6
(C) 49.6
(D) 50.6

Q.18

Find the ratio in which tomatoes worth ₹45/kg can be mixed with tomatoes worth ₹60/kg, so that the mixture will be worth ₹54/kg.

(A) 3 : 5
(B) 2 : 3
(C) 2 : 5
(D) 3 : 7

Q.19

What sum (in ₹) will earn a simple interest of ₹360 in 2 years at 4% per year rate of interest?

(A) 6000
(B) 9000
(C) 2250
(D) 4500

Q.20

A well with an internal diameter of 8 m is dug to a depth of 12 m. The earth taken out of it is spread all around the well to a width of 8 m to form an embankment. What is the height (in m) of the embankment?

(A) 2.2
(B) 1.5
(C) 1.0
(D) 1.3

Reasoning Ability and Problem Solving 15 questions

Q.1

In a certain code language, 'YOUR' is coded as '1245' and 'OURS' is coded as '4216'. What is the code for 'Y' in that language?

(A) 6
(B) 5
(C) 1
(D) 4

Q.3

U, V, W, X, Y, K and A are sitting around a circular table facing the centre (but not necessarily in the same order) . Only two people live between W and X when counted from the left of X.U sits third to the left of K.V sits to the immediate right of K.V sits second to the left of W.Y is not an immediate neighbour of U. Who sits third to the right of A?

(A) V
(B) W
(C) U
(D) K

Q.4

Which numbers should come in place of the question marks (?) in the same sequence to make the series logically complete? 19 ,33, 21, 30 ,23 ,27, ?, 24 ,?

(A) 28, 25
(B) 23, 25
(C) 25, 26
(D) 25, 27

Q.5

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some watches are belts. Some belts are ties. Conclusions: (I) Some watches are ties. (II) Some belts are watches.

(A) Only conclusion II follows
(B) Both conclusion I and II follow
(C) Neither conclusion I nor II follows
(D) Only conclusion I follows

Q.6

In a certain code language, 'PORK' is written as '7358' and 'CREW' is written as '1693'. How is 'R' written in that code language?

(A) 6
(B) 9
(C) 3
(D) 5

Q.7

5 is related to 120 following a certain logic. Following the same logic, 3 is related to 24. To which of the following is 2 related following the same logic?(NOTE: Operations should be performed on the whole numbers, without breaking down the numbersinto its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1and 3 is not allowed)

(A) 11
(B) 7
(C) 6
(D) 13

Q.9

Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (2, 70, 82) (26, 17, 53)

(A) (36, 42, 90)
(B) (79, 12, 141)
(C) (54, 88, 152)
(D) (71, 78, 122)

Q.10

GI23 is related to SU31 in a certain way. In the same way, MO56 is related to YA64. To which of the following is SU78 related, following the same logic?

(A) EG68
(B) EG86
(C) GE66
(D) GE68

Q.11

P, Q, R, S, T and U are sitting around a round table facing the centre (but not necessarily in the same order). P is sitting to the immediate right of U and immediate left of S. S is sitting to the immediate left of T. Q is sitting to the immediate right of T and immediate left of R. Who is sitting to the immediate right of R?

(A) T
(B) P
(C) U
(D) S

Q.12

What will come in place of the question mark (?) in the following equation if '+' and '–' are interchanged and '×' and '÷' are interchanged? $84 - 112 \times 8 + 15 \div 3 = ?$

(A) 63
(B) 57
(C) 53
(D) 45

Q.13

QBON is related to UFSR in a certain way based on the English alphabetical order. In the same way, FCLT is related to JGPX. To which of the following is AOGS related, following the same logic?

(A) ERJV
(B) CQIU
(C) BQJW
(D) ESKW

Q.14

What should come in place of the question mark (?) in the following series based on the English alphabetical order? RSP ,TQN ,VOL, XMJ, ?

(A) AKH
(B) YLH
(C) ZKH
(D) YKH

Q.15

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: No bat is a wicket. No wicket is a glove. Conclusions: (I) No bat is a glove. (II) No wicket is a bat.

(A) Only conclusion I follows.
(B) Both conclusions I and II follow.
(C) Neither conclusion I nor II follows.
(D) Only conclusion II follows.

Q.17

Which two signs should be interchanged to make the given equation correct? 60 × 10 + 12 ÷ 4 – 16 = 74

(A) + and ×
(B) – and +
(C) ÷ and ×
(D) – and ×

Q.20

In a certain code language, 'WOLF' is coded as '4729' and 'FLEW' is coded as '9254'. What is the code for 'E' in that language?

(A) 9
(B) 4
(C) 5
(D) 2

General Awareness 25 questions

Q.1

In which state is the 'slash and burn' agriculture known as 'Dahiya'?

(A) Andhra Pradesh
(B) Himachal Pradesh
(C) Arunachal Pradesh
(D) Madhya Pradesh

Q.2

Which of the following is the principal strategy adopted by National Literacy Mission for the eradication of illiteracy?

(A) Total Literacy Campaign
(B) Sarva Shiksha Abhiyan
(C) Education Guarantee Scheme
(D) Alternative and Innovative Education

Q.3

Which of the following statements is true about Micro Finance institutions?

(A) Majority of microfinancing operations occur in developing countries
(B) Service provided to High income people only
(C) Can provide jumbo loan
(D) Type of banking service provided to employed

Q.4

Which of the following personalities had the key role in fighting against the British East India Company in the Battle of Porto Novo?

(A) Gopala Nayaker
(B) Alamgir-II
(C) Hyder Ali
(D) Maharaja Ranjit Singh

Q.5

In which year did India first participate in the Paralympic Games?

(A) 1964
(B) 1968
(C) 1980
(D) 1972

Q.6

The construction of Humayun's tomb was completed by:

(A) Rahmat Banu Begum
(B) Zeenat Mahal
(C) Mah Chuchak Begum
(D) Biga Begum

Q.7

Select the imbalanced chemical equation from the following.

(A) $10\ KClO_3 + 3\ P_4 \rightarrow 3\ P_4O_{10} + 10\ KCl$
(B) $3\ Hg(OH)_2 + 2\ H_3PO_4 \rightarrow Hg_3(PO_4)_2 + 6\ H_2O$
(C) $6\ HClO_4 + P_4O_{10} \rightarrow 4\ H_3PO_4 + 6\ Cl_2O_7$
(D) $3\ CaCl_2 + 2\ Na_3PO_4 \rightarrow Ca_3(PO_4)_2 + 6\ NaCl$

Q.8

Who built Sudarshan lake in Junagarh?

(A) Chandra Gupta
(B) Skandgupta
(C) Pushyagupta
(D) Bhanugupta

Q.9

Who among the following was appointed as the director-general in the Directorate General of Civil Aviation in February 2023?

(A) Pramod Kumar Mishra
(B) Vikram Dev Dutt
(C) Ajay Kumar Bhalla
(D) Rajiv Gauba

Q.10

Which of the following statements about phylum Arthropoda are correct? A. Animals of Arthropoda have joint appendages. B. They have compounds eyes. C. These make up the majority of all animals on earth. D. The mode of reproduction in all Arthropods is vegetative.

(A) Only B, C and D
(B) Only C and D
(C) Only A, B and C
(D) Only A and C

Q.11

Consider the following statements in the context of the Mughal Rule. 1. The Mansabdari System was introduced by Akbar in 1571. 2. The Second Battle of Panipat was fought in 1556. 3. Maharana Pratap Singh of Mewar was a contemporary of Babur. Which of the statements given above is/are correct?

(A) Both 2 and 3
(B) Both 1 and 2
(C) Only 2
(D) 1, 2 and 3

Q.12

What is the rank of India in the Global Hunger Index (GHI) 2023?

(A) 111
(B) 120
(C) 115
(D) 105

Q.13

In February 2023, UNICEF India announced the appointment of which celebrated Bollywood star as a National Ambassador for Child Rights?

(A) Pankaj Tripathi
(B) Aamir Khan
(C) Ayushmann Khurrana
(D) Pankaj Kapoor

Q.14

Which of the following dance categories is NOT considered for an award of the Sangeet Natak Akademi?

(A) Manipuri
(B) Kathak
(C) Kuchipudi
(D) Vilasini Natyam

Q.15

The Patli Dun is associated with which physiographic division of India?

(A) The Himalayan Mountains
(B) The Indian Desert
(C) The Peninsular Plateau
(D) The Coastal Plains

Q.16

Yashodhara Dalmia is the author of which famous book?

(A) The Painted World of the Warlis
(B) Paintings in the Kangra Valley
(C) Ajanta
(D) Indian Art

Q.17
Manipuri Raas Leela, a renowned form of classical dance, was created by:

(A) Siddhendra Yogi
(B) Sage Bharata
(C) Birju Maharaj
(D) Maharaja Bhagyachandra

Q.18
Which country hosted the Football World Cup, 2022?

(A) France
(B) Argentina
(C) Qatar
(D) UAE

Q.19
Which of the following does NOT figure among the themes of 'Azadi Ka Amrit Mahotsav'?

(A) Resolve@75
(B) Ideas@75
(C) Health@75
(D) Actions@75

Q.20
What is the primary function of the notochord in chordates?

(A) Support
(B) Locomotion
(C) Digestion
(D) Respiration

Q.21
Fuels that are derived from fossils are preserved only in which kind of rocks?

(A) Metamorphic rocks
(B) Plutonic rocks
(C) Volcanic rocks
(D) Sedimentary rocks

Q.22
Which Article of the Indian Constitution deals with the appointment and duties of the Attorney-General of India?

(A) Article 74
(B) Article 81
(C) Article 80
(D) Article 76

Q.23
Which of following organisations sought to revive Vedic learning and combine it with modern education in the sciences?

(A) Arya Samaj
(B) Satyashodhak Samaj
(C) Brahmo Samaj
(D) Prarthana Samaj

Q.24
Which Article states that it shall be the duty of every citizen of India to abide by the constitution and respect its ideals and institutions, the National Flag and the National Anthem?

(A) 52
(B) 14
(C) 51A (a)
(D) 54 (b)

Q.25
Who explained the system of classifying musical instruments into Tata, Sushira, Avanaddha and Ghana in his treatise 'Natya Shastra' in the early 2nd century?

(A) Swami Haridas
(B) Bhairav
(C) Sarngadeva
(D) Bharata Muni

English Language and Comprehension 23 questions

Q.1

Select the most appropriate option to fill in the blank. As 'brutal' is synonymous to 'harsh' , 'naive' is synonymous to _____.

(A) simple
(B) worldly
(C) wise
(D) hospitable

Q.2

Select the option that will improve the underlined part of the given sentence. In case no improvement is needed, select 'No improvement required'. Mukta "requested Nikita to share" the picture taken during the wedding.

(A) No improvement required
(B) requesting Nikita to share
(C) will requests Nikita to share
(D) requested Nikita for share

Q.3

Select the appropriate option that can substitute the underlined word in the following sentence to make it grammatically correct. The grapes taste "sweetest".

(A) sweeter
(B) more sweeter
(C) most sweeter
(D) sweet

Q.4

Select the INCORRECTLY spelt word. Nylon was among the first synthitic fibre made by man and is known for its toughness, strength and resistance to oil and grease.

(A) resistance
(B) toughness
(C) synthitic
(D) strength

Q.6

Select the INCORRECTLY spelt word.

(A) Believe
(B) Thief
(C) Relief
(D) Recieve

Q.7

Select the most appropriate option that can substitute the underlined segment in the given sentence. The intersection of nature conservation and social justice is "evident in efforts to ensure" that marginalised communities and other backward classes should have equal access to natural reserves.

(A) obvious in attempts to recollect
(B) apparent in endeavours to guarantee
(C) Manifest in actions to redesign
(D) Unclear in initiatives to safeguard

Q.8

Select the most appropriate meaning of the given idiom. Hornet's nest

(A) A bee's nest
(B) A house built with great difficulty
(C) A difficult task
(D) A troublesome or hazardous situation

Q.9

The following sentence has been divided into parts. One of them may contain an error. Select the part that contains the error from the given options. If you don't find any error, mark 'No error' as your answer. You have spoilt / the little chance you / had at succeeding.

(A) No error
(B) had at succeeding.
(C) You have spoilt
(D) the little chance you

Q.10

Select the most appropriate option to fill in the blank. Tom will _____ the document to the office tomorrow.

(A) meal
(B) mall
(C) mail
(D) male

Q.11

Select the correctly spelt word to fill in the blank. The play was shortened by the _____ of two scenes.

(A) addition
(B) insirtion
(C) omission
(D) ommision

Q.12

Select the most appropriate option to fill in the blank. I will not be able to _____ my assignment in time.

(A) completing
(B) completed
(C) completes
(D) complete

Q.13

Select the option that will improve the underlined part of the given sentence. In case no improvement is needed, select 'No improvement'. The name, Khajuraho, has probably “been derived from” the word ‘Khajoor’, meaning date palm.

(A) derives into
(B) be derived into
(C) derive at
(D) No improvement

Q.14

The following sentence contains an error. Select the option that correctly rectifies the error. The robbers stabbed him by a sharp knife and he started bleeding profusely on the spot.

(A) profusely from the spot
(B) The robbers stabbed her
(C) with a sharp knife
(D) but he started bleeding

Q.16

The following sentence has been split into four segments. Identify the segment that contains an error. That question is / much difficult / but this one is / very easy.

(A) very easy.
(B) but thisone is
(C) much difficult
(D) That question is

Q.17

Select the most appropriate option that can substitute the underlined segment in the given sentence. The manager complained that Sheetal's email was not “up to the standard” for the company.

(A) under the nose
(B) up to the mark
(C) up and doing
(D) an uphill task

Q.18

Select the word that has the same meaning (SYNONYM) as the underlined word in the given sentence. The vital role played by women at home and in the struggle for freedom, advanced the case for female “emancipation” soon after the war was over.

(A) objectivity
(B) bondage
(C) utility
(D) freedom

Q.19

Identify the most appropriate ANTONYM of the word in brackets. The magician's performance was (captivating) and held the audience spellbound.

(A) Enchanting
(B) Gripping
(C) Boring
(D) Fascinating

Q.20

Select the most appropriate synonym of the underlined word. Her “amicable” nature made it easy for her to forge positive relationships.

(A) Friendly
(B) Attentive
(C) Sad
(D) Aggressive

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Saltwater crocodiles are also known as estuarine crocodiles, as they (1)_____ to live in mangrove-lined rivers. They’re the (2)_____ living reptile, reaching up to seven metres in length: far larger than Indonesia’s famous Komodo dragon, which tops out at three metres. (3)_____, crocodiles lived throughout the Indonesian archipelago. We have (4)_____ of attacks on humans in Bali from the early 20th century and across much of Java until the 1950s. Even Indonesia’s capital, Jakarta, had crocodile’s (5)_____in many rivers running through the city.

Select the most appropriate option to fill in blank no. 1.

(A) defer
(B) refer
(C) prefer
(D) differ

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Saltwater crocodiles are also known as estuarine crocodiles, as they (1)_____ to live in mangrove-lined rivers. They’re the (2)_____ living reptile, reaching up to seven metres in length: far larger than Indonesia’s famous Komodo dragon, which tops out at three metres. (3)_____, crocodiles lived throughout the Indonesian archipelago. We have (4)_____ of attacks on humans in Bali from the early 20th century and across much of Java until the 1950s. Even Indonesia’s capital, Jakarta, had crocodile’s (5)_____in many rivers running through the city.

Select the most appropriate option to fill in blank no. 2.

(A) broadest
(B) fattest
(C) poorest
(D) largest

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Saltwater crocodiles are also known as estuarine crocodiles, as they (1)_____ to live in mangrove-lined rivers. They’re the (2)_____ living reptile, reaching up to seven metres in length: far larger than Indonesia’s famous Komodo dragon, which tops out at three metres. (3)_____, crocodiles lived throughout the Indonesian archipelago. We have (4)_____ of attacks on humans in Bali from the early 20th century and across much of Java until the 1950s. Even Indonesia’s capital, Jakarta, had crocodile’s (5)_____in many rivers running through the city.

Select the most appropriate option to fill in blank no. 3.

(A) Abysmally
(B) Historically
(C) Fanatically
(D) acrobatically

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Saltwater crocodiles are also known as estuarine crocodiles, as they (1)_____ to live in mangrove-lined rivers. They're the (2)_____ living reptile, reaching up to seven metres in length: far larger than Indonesia's famous Komodo dragon, which tops out at three metres. (3)_____, crocodiles lived throughout the Indonesian archipelago. We have (4)_____ of attacks on humans in Bali from the early 20th century and across much of Java until the 1950s. Even Indonesia's capital, Jakarta, had crocodile's (5)_____in many rivers running through the city.

Select the most appropriate option to fill in blank no. 4.

(A) recipes
(B) residues
(C) regiments
(D) records

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Saltwater crocodiles are also known as estuarine crocodiles, as they (1)_____ to live in mangrove-lined rivers. They're the (2)_____ living reptile, reaching up to seven metres in length: far larger than Indonesia's famous Komodo dragon, which tops out at three metres. (3)_____, crocodiles lived throughout the Indonesian archipelago. We have (4)_____ of attacks on humans in Bali from the early 20th century and across much of Java until the 1950s. Even Indonesia's capital, Jakarta, had crocodile's (5)_____in many rivers running through the city.

Select the most appropriate option to fill in blank no. 5.

(A) remembrance
(B) relevance
(C) residence
(D) resistance

Answer key

Numerical and Mathematical Ability

Q.1 **(D)**	Q.3 **(B)**	Q.4 **(D)**	Q.7 **(B)**	Q.8 **(A)**	Q.9 **(B)**
Q.10 **(A)**	Q.12 **(B)**	Q.13 **(A)**	Q.14 **(A)**	Q.15 **(D)**	Q.18 **(B)**
Q.19 **(D)**	Q.20 **(B)**				

Reasoning Ability and Problem Solving

Q.1 **(B)**	Q.3 **(A)**	Q.4 **(D)**	Q.5 **(A)**	Q.6 **(C)**	Q.7 **(C)**
Q.9 **(C)**	Q.10 **(B)**	Q.11 **(C)**	Q.12 **(C)**	Q.13 **(D)**	Q.14 **(C)**
Q.15 **(D)**	Q.17 **(A)**	Q.20 **(C)**			

General Awareness

Q.1 **(D)**	Q.2 **(A)**	Q.3 **(A)**	Q.4 **(C)**	Q.5 **(B)**	Q.6 **(D)**
Q.7 **(C)**	Q.8 **(C)**	Q.9 **(B)**	Q.10 **(C)**	Q.11 **(B)**	Q.12 **(A)**
Q.13 **(C)**	Q.14 **(D)**	Q.15 **(A)**	Q.16 **(A)**	Q.17 **(D)**	Q.18 **(C)**
Q.19 **(C)**	Q.20 **(A)**	Q.21 **(D)**	Q.22 **(D)**	Q.23 **(A)**	Q.24 **(C)**
Q.25 **(D)**					

English Language and Comprehension

Q.1 **(A)**	Q.2 **(A)**	Q.3 **(D)**	Q.4 **(C)**	Q.6 **(D)**	Q.7 **(B)**
Q.8 **(D)**	Q.9 **(A)**	Q.10 **(C)**	Q.11 **(C)**	Q.12 **(D)**	Q.13 **(D)**
Q.14 **(C)**	Q.16 **(C)**	Q.17 **(B)**	Q.18 **(D)**	Q.19 **(C)**	Q.20 **(A)**
Q.21 **(C)**	Q.22 **(D)**	Q.23 **(B)**	Q.24 **(D)**	Q.25 **(C)**	

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (D) 82

If the average before the 15th inning was a, then 14a + 152 = 15(a + 5), so a = 77 and the new average is 82.

Q.3 · Answer (B) ₹24,720

The amounts collected are in the ratio (5 × 3) : (3 × 10) = 15 : 30 = 1 : 2, so second class brought in 2/3 × ₹37,080 = ₹24,720.

Q.4 · Answer (D) 4,000

₹2,108 is 0.62 × 0.85 = 0.527 of the marked price, so the marked price is 2,108/0.527 = ₹4,000.

Q.7 · Answer (B) 10

Inside the first bracket: 25 + 5 − 25 + 5 − 25 = −15. Then, left to right, (5 × 5) ÷ 5 × (2 + 3) = 25 ÷ 5 × 5 = 25. −15 + 25 = 10.

Q.8 · Answer (A) 3,800

The interest comes to 4% × 2 + 8% × 3 + 12% × 1 = 44% of the sum, so 1.44P = ₹5,472 and P = ₹3,800.

Q.9 · Answer (B) 78.57%

He gives 0.7 kg for the price of 1 kg, which is 1.25 times its cost. Profit = 1.25/0.7 − 1 = 0.7857, or 78.57%.

Q.10 · Answer (A) 72 km/h

The train covers 735 + 225 = 960 m in 48 s: 20 m/s, or 20 × 18/5 = 72 km/h.

Q.12 · Answer (B) 3 : 2

For a 10% profit the mixture must cost 34.1/1.1 = ₹31 a kg. By alligation the ratio is (32.5 − 31) : (31 − 30) = 1.5 : 1 = 3 : 2.

Q.13 · Answer (A) 16,380

1.25R + 1.1G = 30,400 and 1.1R + 1.25G = 30,700. Adding: 2.35(R + G) = 61,100, so R + G = 26,000; subtracting: 0.15(G − R) = 300, so G − R = 2,000. The Geyser costs ₹14,000, and at 17% profit sells for 14,000 × 1.17 = ₹16,380.

Q.14 · Answer (A) 960

SI = 3,000 × 8 × 4/100 = ₹960.

Q.15 · Answer (D) 50.6

(40 + 49 + 56 + 71 + 37)/5 = 253/5 = 50.6 kg.

Q.18 · Answer (B) 2 : 3

By alligation: (60 − 54) : (54 − 45) = 6 : 9 = 2 : 3.

Q.19 · Answer (D) 4500

P = SI × 100/(R × T) = 360 × 100/(4 × 2) = ₹4,500.

Q.20 · Answer (B) 1.5

Earth dug out = π × 4^2 × 12 = 192π m^3. The embankment is a ring from radius 4 m to 12 m, of area π(12^2 − 4^2) = 128π m^2, so its height is 192π/128π = 1.5 m.

Reasoning Ability and Problem Solving

Q.1 · Answer (B) 5

O, U and R are in both words, as are 1, 2 and 4 in both codes. The letter left in YOUR is Y and the digit left in its code is 5, so Y = 5.

Q.3 · Answer (A) V

Going round to the left from K the order is K, X, A, U, W, Y, V: V to K's immediate right, U third to K's left, W second to V's right, and X with two people between it and W. Y cannot sit next to U, so it takes the seat between W and V, leaving A between X and U. Third to the right of A is V.

Q.4 · Answer (D) 25, 27

The terms in odd places rise by 2 (19, 21, 23, 25, 27) and those in even places fall by 3 (33, 30, 27, 24). The missing 7th and 9th terms are 25 and 27.

Q.5 · Answer (A) Only conclusion II follows

'Some watches are belts' converts to 'some belts are watches', so II follows. The belts that are ties need not be watches, so I does not follow.

Q.6 · Answer (C) 3

R is the only letter in both words and 3 the only digit in both codes, so R = 3.

Q.7 · Answer (C) 6

Each number is related to $n^3 - n$: $5^3 - 5 = 120$ and $3^3 - 3 = 24$. Likewise $2^3 - 2 = 6$.

Q.9 · Answer (C) (54, 88, 152)

The third number is the sum of the first two plus 10: 2 + 70 + 10 = 82 and 26 + 17 + 10 = 53. Likewise 54 + 88 + 10 = 152.

Q.10 · Answer (B) EG86

Each letter moves 12 places forward (wrapping round after Z) and the number rises by 8: GI23 → SU31 and MO56 → YA64. So SU78 → EG86.

Q.11 · Answer (C) U

Going round to the right from U the order is U, P, S, T, Q, R, and the circle then closes at U, so U sits to the immediate right of R.

Q.12 · Answer (C) 53

After interchanging the signs: 84 + 112 ÷ 8 – 15 × 3 = 84 + 14 – 45 = 53.

Q.13 · Answer (D) ESKW

Each letter moves 4 places forward: QBON → UFSR and FCLT → JGPX. So AOGS → ESKW.

Q.14 · Answer (C) ZKH

The first letters move +2 (R, T, V, X, Z), the second –2 (S, Q, O, M, K) and the third –2 (P, N, L, J, H): ZKH.

Q.15 · Answer (D) Only conclusion II follows.

'No bat is a wicket' converts to 'no wicket is a bat', so II follows. Two negative statements give no conclusion about bats and gloves, so I does not follow.

Q.17 · Answer (A) + and ×

Interchanging + and ×: 60 + 10 × 12 ÷ 4 – 16 = 60 + 30 – 16 = 74.

Q.20 · Answer (C) 5

W, L and F are in both words, as are 4, 2 and 9 in both codes. The letter left in FLEW is E and the digit left in its code is 5, so E = 5.

General Awareness

Q.1 · Answer (D) Madhya Pradesh

NCERT lists 'Dahiya' as Madhya Pradesh's name for slash-and-burn (shifting) cultivation; it is 'Podu' or 'Penda' in Andhra Pradesh and 'Jhumming' in the north-east.

Q.2 · Answer (A) Total Literacy Campaign

The National Literacy Mission, launched in 1988, adopted the Total Literacy Campaign, an area-specific, time-bound, volunteer-based drive, as its principal strategy.

Q.3 · Answer (A) Majority of microfinancing operations occur in developing countries
Microfinance offers small loans and other financial services to low-income people who lack access to ordinary banking, and most of it operates in developing countries.

Q.4 · Answer (C) Hyder Ali
At Porto Novo in 1781, during the Second Anglo-Mysore War, Hyder Ali of Mysore fought the East India Company's army under Sir Eyre Coote, and was defeated.

Q.5 · Answer (B) 1968
India first took part in the Paralympic Games at Tel Aviv in 1968; its first medal, a gold for the swimmer Murlikant Petkar, came in 1972.

Q.6 · Answer (D) Biga Begum
Humayun's tomb in Delhi was built by his widow Bega (Biga) Begum, also called Haji Begum, to a design by Mirak Mirza Ghiyas.

Q.7 · Answer (C) 6 $HClO_4 + P_4O_{10} \rightarrow 4\ H_3PO_4 + 6\ Cl_2O_7$
In 6 $HClO_4 + P_4O_{10} \rightarrow 4\ H_3PO_4 + 6\ Cl_2O_7$ the left side has 6 hydrogen and 6 chlorine atoms but the right has 12 of each; it balances with 12 $HClO_4$. The other three have the same number of atoms of each element on both sides.

Q.8 · Answer (C) Pushyagupta
Rudradaman's Junagadh inscription records that the Sudarshan lake was built by Pushyagupta, Chandragupta Maurya's governor in Saurashtra; it was repaired later by Rudradaman and again under Skandagupta.

Q.9 · Answer (B) Vikram Dev Dutt
Vikram Dev Dutt, an IAS officer, was appointed Director General of Civil Aviation in February 2023.

Q.10 · Answer (C) Only A, B and C
Arthropods have jointed appendages, many have compound eyes, and they make up most of the known animal species. D is wrong: vegetative reproduction is a mode of plants, not of animals.

Q.11 · Answer (B) Both 1 and 2
SSC's key treats 1 and 2 as correct: Akbar introduced the mansabdari system (1571 is the year exam sources usually give, though historians date its full form later), and the Second Battle of Panipat was fought in 1556. 3 is wrong: Maharana Pratap, born in 1540, was Akbar's contemporary; Babur died in 1530.

Q.12 · Answer (A) 111
India ranked 111th of 125 countries in the Global Hunger Index 2023.

Q.13 · Answer (C) Ayushmann Khurrana
In February 2023 UNICEF India named Ayushmann Khurrana its National Ambassador for child rights.

Q.14 · Answer (D) Vilasini Natyam
SSC's key gives Vilasini Natyam, a temple and court dance tradition of Andhra Pradesh, which is not among the eight classical dance forms the Sangeet Natak Akademi recognises; Manipuri, Kathak and Kuchipudi are.

Q.15 · Answer (A) The Himalayan Mountains
The Duns are longitudinal valleys of the Himalayas, lying between the Lesser Himalaya and the Shiwaliks; Dehra Dun, Kotli Dun and Patli Dun are well-known ones.

Q.16 · Answer (A) The Painted World of the Warlis
The art historian Yashodhara Dalmia wrote 'The Painted World of the Warlis', on the art and ritual of the Warli tribe of Maharashtra.

Q.17 · Answer (D) Maharaja Bhagyachandra
Manipuri Raas Leela was created by Maharaja Bhagyachandra (Rajarshi Bhagyachandra) of Manipur in the 18th century.

Q.18 · Answer (C) Qatar
The 2022 FIFA World Cup was held in Qatar, the first in the Arab world; Argentina won it.

Q.19 · Answer (C) Health@75
Azadi Ka Amrit Mahotsav has five themes: Freedom Struggle, Ideas@75, Resolve@75, Actions@75 and Achievements@75. Health@75 is not one of them.

Q.20 · Answer (A) Support
The notochord is a flexible rod along the back that supports the body; in vertebrates the backbone replaces it.

Q.21 · Answer (D) Sedimentary rocks
Coal, petroleum and natural gas form from organic matter buried in layers of sediment, so they are found in sedimentary rocks.

Q.22 · Answer (D) Article 76
Article 76 provides for the Attorney-General for India, appointed by the President, and sets out the office's duties.

Q.23 · Answer (A) Arya Samaj
The Arya Samaj, founded by Swami Dayanand Saraswati in 1875, sought to revive Vedic learning and combine it with modern scientific education, as in its DAV schools.

Q.24 · Answer (C) 51A (a)
Article 51A lists the Fundamental Duties, and this is the first of them, clause (a).

Q.25 · Answer (D) Bharata Muni
Bharata Muni's 'Natya Shastra' divides instruments into tata (stringed), sushira (wind), avanaddha (skin-covered drums) and ghana (solid instruments, such as cymbals).

English Language and Comprehension

Q.1 · Answer (A) simple
'Brutal' and 'harsh' are alike in meaning; 'naive' means innocent and unworldly: 'simple'. 'Worldly' and 'wise' are nearer its opposite.

Q.2 · Answer (A) No improvement required
'Request someone to do something' is correct, so the sentence needs no change.

Q.3 · Answer (D) sweet
'Taste' is a linking verb followed by an adjective, and with no comparison the positive degree is needed: 'The grapes taste sweet.'

Q.4 · Answer (C) synthitic
The correct spelling is 'synthetic'.

Q.6 · Answer (D) Recieve
The correct spelling is 'receive': 'i' before 'e' except after 'c'.

Q.7 · Answer (B) apparent in endeavours to guarantee
'Evident' means 'apparent', 'efforts' are 'endeavours' and to 'ensure' is to 'guarantee', so the meaning is kept; the other options change it.

Q.8 · Answer (D) A troublesome or hazardous situation
A 'hornet's nest' is a troublesome or dangerous situation, as in 'stir up a hornet's nest'.

Q.9 · Answer (A) No error
SSC's key gives no error: 'the little chance you had' means what small chance there was, and 'a chance at' something is accepted usage, though 'a chance of succeeding' is the more usual form.

Q.10 · Answer (C) mail
To 'mail' a document is to send it by post; 'meal', 'mall' and 'male' only sound alike.

Q.11 · Answer (C) omission
Leaving out two scenes, an 'omission', shortens a play; 'ommision' and 'insirtion' are misspelt.

Q.12 · Answer (D) complete
'Be able to' is followed by the base form of the verb: 'able to complete'.

Q.13 · Answer (D) No improvement
A word is 'derived from' another, so the underlined part is correct as it stands.

Q.14 · Answer (C) with a sharp knife
The instrument used takes 'with': 'stabbed him with a sharp knife'.

Q.16 · Answer (C) much difficult
'Much' qualifies comparatives ('much more difficult'); with the positive degree use 'very': 'very difficult'.

Q.17 · Answer (B) up to the mark
'Up to the mark' means of the required standard.

Q.18 · Answer (D) freedom
'Emancipation' is being set free from legal, social or political restrictions: 'freedom'. 'Bondage' is its opposite.

Q.19 · Answer (C) Boring
'Captivating' means holding the attention; its opposite is 'boring'. The other three are its synonyms.

Q.20 · Answer (A) Friendly
'Amicable' means friendly and peaceable.

Q.21 · Answer (C) prefer
Saltwater crocodiles 'prefer' to live in mangrove-lined rivers, which is why they are also called estuarine crocodiles.

Q.22 · Answer (D) largest
Reaching up to seven metres, far larger than the Komodo dragon, they are the 'largest' living reptile.

Q.23 · Answer (B) Historically
'Historically', that is in the past, crocodiles lived throughout the archipelago; the sentences that follow describe the past.

Q.24 · Answer (D) records
'We have records of attacks': written accounts from the early 20th century to the 1950s.

Q.25 · Answer (C) residence
Jakarta had crocodiles in 'residence', that is living, in many of the rivers running through the city.

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